

Short-Horizon Electricity Forecasting for a Steel Plant

ML System Design Document

Assignment	Individual Assignment 1
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Executive summary

This report designs a decision-support system that forecasts electricity consumption for a small steel plant and warns operators about upcoming high-load intervals. It uses the public UCI Steel Industry Energy Consumption dataset: 35,040 records from 2018, attributed to a plant in South Korea. A local audit confirms the file size, completeness and a recurring midnight timestamp issue. The design explicitly separates those observations from assumptions about future deployment. A single tabular regression model uses completed energy measurements and known calendar information; contemporaneous electrical measurements and load labels are excluded to prevent target leakage. Forecasts support a review of deferrable auxiliary loads, while production and electrical protection remain under existing controls. The architecture combines scheduled ingestion, time-aware feature generation, MLflow model lineage, Airflow orchestration and a deterministic fallback. Model accuracy is evaluated chronologically, and business benefit requires a separate prospective pilot. This assignment establishes the system design and data feasibility; it does not claim trained-model performance, achieved savings or a deployed system.

Scope: one aggregate plant meter, one regression model, 15-minute energy intervals and operator-reviewed load alerts. Public data are verified; deployment conditions and acceptance targets are proposed.

1 PROBLEM DEFINITION | 20%

An operational forecasting problem

1.1 Context and current situation

Steel production combines sustained electrical demand with abrupt changes in operating activity. A short-horizon forecast can help an energy manager identify an approaching high-load interval and ask an operator whether a non-critical auxiliary activity can be deferred. The public dataset describes DAEWOO Steel Co. Ltd in Gwangyang, South Korea [1]. This report proposes a deployment scenario inspired by that dataset; it does not claim access to that company's current procedures, tariff or equipment controls.

The assumed current process is a meter dashboard with retrospective consumption charts and fixed threshold alarms. Operators use experience to judge whether an upcoming auxiliary start can wait. A warning after a peak begins offers less preparation time. The actual plant workflow, controllable equipment and response time must be confirmed before a pilot.

1.2 Problem statement and boundaries

Every 15 minutes, use information available at issue time to predict energy in a specified future 15-minute interval. Provide a non-negative kWh forecast and a versioned high-load advisory. The offline goal is at least 10% lower MAE than the strongest eligible simple baseline, with no deterioration in high-load MAE. The prospective business target is a 5% reduction in daily peak 15-minute average demand on comparable operating days, with no forecast-induced production interruption. These are acceptance proposals, not results.

The first release covers one aggregate meter and one regression target. It excludes automatic machine control, equipment-level allocation, tariff optimisation, fault protection, carbon accounting and day-ahead scheduling. An advisory does not establish electrical safety or guarantee a peak will occur.

Stakeholder	Goal and concern	Decision responsibility
Energy manager	Anticipate high demand without excessive false alarms.	Owns advisory settings and reviews daily energy outcomes.
Production / shift operator	Protect throughput, product quality and safe sequences.	Accepts or rejects suggested review of deferrable loads; records reasons.
Finance / plant management	Verify net benefit against implementation and operating costs.	Approves pilot budget and any eventual business case.
IT / data team	Maintain timely, reproducible and secure forecasts.	Owns data contracts, releases, monitoring, fallback and recovery.

1.3 Why machine learning is appropriate

Demand can depend on recent consumption, time of day, weekday and interactions among these signals. A small tree model can capture nonlinear transitions without a specialised deep-learning platform. However, persistence and seasonal forecasts may already be strong at this horizon. ML is justified only if an honest future-period evaluation beats them; otherwise the baseline remains the operational choice.

2 REQUIREMENTS ANALYSIS | 20%

Define the forecast before the model

2.1 Decision timing and prediction contract

Let t be the end of the latest completed 15-minute meter interval. Assume that reading arrives within 30 seconds and the forecast is published by $t + 60$ seconds. Predict energy for $[t + 15 \text{ minutes}, t + 30 \text{ minutes})$: the next interval after the immediately upcoming one. This provides at least 14 minutes before the target interval starts. The horizon is 30 minutes to the target interval end; it is not a prediction of energy already observed.



Target interval length: 15 minutes. Publication leaves 14 minutes before its start.

Feasibility gate: operators must confirm that a safe, optional auxiliary-load decision is possible within 14 minutes. If that lead time is insufficient, the horizon must be extended and evaluated again; model accuracy alone does not validate the workflow.

ID	Functional requirement	Acceptance evidence
FR-01	Ingest meter readings with interval boundaries, arrival time, units and source ID.	An immutable source record and quality status exist for each reading.
FR-02	Generate one forecast per meter and target interval using a frozen input snapshot.	Output includes target window, kWh, issue time, data cutoff and model version.
FR-03	Apply a separately versioned alert rule; display forecast age and quality.	An advisory is distinguishable from stale, fallback or unavailable output.
FR-04	Allow operator acknowledgement, rejection and reason-coded action notes.	Decisions are audited; no endpoint directly commands a machine.
FR-05	Serve the latest valid forecast through an internal API and dashboard.	Idempotent writes cannot create duplicate forecasts or duplicate alerts.
FR-06	Join each forecast with the later meter outcome and monitor accuracy.	Predictions remain immutable; corrections create a new labelled revision.

Illustrative contract only

```
latest_interval_end=10:00; issued_at=10:01; target=[10:15,10:30)
forecast_kwh=30.0; average_kw=120.0; quality=VALID
meter_id=plant01; model_version=3; alert_policy_version=2
```

All times in this example refer to the plant's local clock. The production contract stores timezone-aware timestamps in UTC and displays Asia/Seoul. Assuming energy is an interval total, average kW = kWh / 0.25 h = 4 x kWh. This conversion describes a 15-minute average, not instantaneous electrical power.

2 REQUIREMENTS ANALYSIS | 20%

Operational, scale and access requirements

ID	Requirement	Proposed target and verification
NFR-01	Forecast timeliness	At least 99% of all scheduled forecasts published within 60 s of t over 30 days. Missing readings count as missed service; report the cause separately.
NFR-02	Response and throughput	Internal API p95 below 500 ms at 10 concurrent readers. Support 96 forecasts/day for one meter; load-test a future 100-meter, 9,600/day deployment.
NFR-03	Reliability and graceful degradation	99.5% monthly API availability. If the ML path fails, use a validated deterministic fallback; if its required data also fail, publish UNAVAILABLE.
NFR-04	Recovery and maintenance	Rollback to a validated model bundle within 15 minutes. Audit-store RPO at most 24 hours and RTO at most 4 hours; test restoration quarterly.
NFR-05	Maintainability and cost	CPU-only inference; weekly candidate training, daily outcome checks and monthly release review. Initial single-site operating budget: USD 100/month, to be estimated and approved.
NFR-06	Security and accountability	TLS, least-privilege access, encrypted backups and secrets outside Git. Separate operator, reviewer and administrator roles; audit access and policy changes.

2.2 Integration and quality gates

Historical CSV enables offline replay; live deployment requires a read-only gateway supplying interval energy and arrival timestamps. Validate schema, unique meter/interval keys, finite non-negative energy, units, cadence and freshness. Quarantine conflicting duplicates. Missing critical lags block ML; never replace missing data with zero or fill from future observations.

Store forecasts and acknowledgements in PostgreSQL and immutable snapshots/models in object storage. FastAPI serves the internal UI. Keep the meter adapter separate from equipment control; a dashboard failure must not interrupt plant operations. Each new site requires local model validation.

For 100 meters, partition a durable work queue by meter_id, preserve per-meter ordering and deduplicate by (meter_id, target_start). Add memory-bounded CPU inference workers with cached approved models; keep training separate. Monitor queue depth, oldest-task age, p95 processing time and deadline misses. Load-test bursts of 100 jobs every 15 minutes. Increase workers when queueing plus processing threatens the 60-second deadline; expire late jobs and use fallback or UNAVAILABLE. This is a proposed design, not measured capacity.

2.3 Privacy, confidentiality and retention

The published file has no employee identifiers. Live plant energy traces can still reveal operating schedules and production patterns. Keep industrial telemetry internal, remove personal identifiers from free-text notes and restrict access by role. Proposed retention is 24 months for raw readings and 36 months for decisions and release manifests, subject to company policy. Keep the UCI attribution and CC BY 4.0 notice with redistributed benchmark data [1, 2].

All service-level thresholds, budgets and retention periods are design assumptions. The public dataset contains no live arrival times, operator actions, production volumes or tariff records; those sources must be added for a prospective pilot.

2 REQUIREMENTS ANALYSIS | 20%

Real data, explicit information boundaries

2.4 Source and locally verified observations

The UCI Steel Industry Energy Consumption dataset is attributed to Sathishkumar V E, Changsun Shin and Yongyun Cho, with DOI 10.24432/C52G8C [1]. The raw CSV was downloaded on 24 September 2026. Its 35,040 rows and 11 columns cover 365 groups of 96 readings. A local audit found zero empty cells, zero duplicate literal timestamps, one zero-energy value and a Usage_kWh range of 0.00 to 157.18. These are file observations, not modelling results.

2.5 Timestamp convention is a versioned assumption

In source-file order, each day runs from 00:15 through 23:45 and ends with 00:00 carrying that same date. This creates 365 backward time steps if timestamps are parsed literally. Preserve the raw order and raw date. For the primary benchmark, interpret each trailing midnight as 24:00 of that source day, then subtract 15 minutes to obtain the interval start. The resulting assumed interval-end grid runs from 2018-01-01 00:15 to 2019-01-01 00:00, with uniform 15-minute gaps.

The file pattern supports this convention but does not independently prove the meter semantics. Record it as timestamp_policy=v1, regenerate calendar features from the normalised target interval start, and repeat evaluation using literal timestamps sorted chronologically as a sensitivity check. Confirm interval boundaries, timezone and kWh aggregation with the meter owner before deployment. If the ranking changes materially, report the ambiguity instead of claiming a robust winner.

Data element	Use in this design	Availability rule
Usage_kWh	Future interval energy is the regression target; older readings are inputs.	For target row j, use only energy rows j-2 or earlier. Row j-1 is not yet complete at issue time.
Calendar fields	Recompute target slot, weekday and cyclical time encodings.	Known in advance; do not reuse inconsistent raw calendar labels after time normalisation.
Reactive energy, power factors and CO2(tCO2)	Excluded from the first-release feature set.	Same-target measurements are unavailable before prediction and may act as target proxies.
Load_Type	Excluded from predictors and target definition.	UCI metadata marks it as the target. This report deliberately defines a separate energy-forecasting task.

Candidate features are energy lags 2, 3, 4, 5, 6, 9, 17, 96 and 672, plus past-only rolling means over 4, 16 and 96 intervals ending at j-2. A 672-interval weekly lag removes the first seven days from common evaluation eligibility: at most 34,368 feature-ready rows remain before other quality exclusions. Fit scalers or other learned transforms only on the relevant training fold.

3 GOALS AND METRICS | 20%

Measure prediction and operational value

The goal hierarchy links forecast skill to usable alerts and then to an independently verified operating benefit. Every threshold below is provisional and must be agreed before final testing. A smaller error is not itself evidence of energy savings, lower bills or improved electrical safety.

Level / owner	Metric and denominator	Acceptance proposal
Business / energy manager	Daily peak 15-minute average demand = $\max(4 \times \text{interval kWh})$ on a local operating day.	5% lower mean daily peak on comparable pilot days versus the approved current process; assess production volume and mix.
Business / production	Forecast-induced interruptions; proportion of peak advisories acknowledged before target start.	Zero attributable interruptions; at least 90% timely acknowledgement in staffed pilot periods.
Business / finance	Net benefit = verified avoided energy/demand charges minus incremental operating cost.	Positive over the pilot; apply the actual tariff and include one-off cost separately. Not estimable from UCI alone.
System / IT	On-time forecasts / all scheduled intervals; monthly API availability; API p95 latency.	At least 99%; at least 99.5%; below 500 ms, respectively.
Model / data owner	MAE (kWh per interval), measured on a common eligible set.	At most 5 kWh and at least 10% lower than the strongest eligible baseline on locked test.
Model / energy manager	High-load MAE; advisory recall and precision for a frozen high-load threshold.	High-load MAE no worse than baseline; recall at least 80% and precision at least 60%. Report support counts.

3.1 Metric definitions and secondary diagnostics

$$\text{MAE} = \sum(|y - \text{prediction}|) / n$$

$$\text{RMSE} = \sqrt{\sum((y - \text{prediction})^2) / n}$$

$$\text{WAPE} = 100 \times \sum(|y - \text{prediction}|) / \sum(y)$$

$$\text{Relative MAE improvement} = 100 \times (\text{MAE}_{\text{baseline}} - \text{MAE}_{\text{model}}) / \text{MAE}_{\text{baseline}}$$

MAE is the primary selection metric because its unit is directly interpretable. Report RMSE for large misses, WAPE for aggregate relative error, signed bias and results by month, weekday/weekend and time of day. Use MAE when total energy is zero; mark WAPE undefined. Avoid per-row MAPE because the raw file includes zero and near-zero demand [3]. If baseline MAE is zero, relative improvement is undefined and ML has no error advantage.

For offline advisory analysis, define high load using the 90th percentile of development-period energy, frozen before calibration. This is a benchmark threshold, not a contractual power limit. Alert when forecast plus a calibrated non-negative margin exceeds that threshold. Recall = detected high-load intervals / all high-load intervals; precision = true high-load alerts / all alerts. Undefined denominators are reported explicitly, never replaced by a perfect score.

3 GOALS AND METRICS | 20%

Chronological evaluation and release gates

3.2 Baselines and fair comparisons

Compare three deterministic baselines: the latest completed energy reading (target lag 2), the corresponding interval one day earlier (lag 96) and one week earlier (lag 672). Select the strongest baseline by pooled development-fold MAE before opening test results. Score candidates and baselines on the same eligible timestamps, using the same issue-time information. Keep operational fallback coverage separate from model accuracy on valid rows.

Window, by target interval start	Purpose	Boundary rule
January-June 2018	Initial fitting window.	Drop warm-up rows; retain only training labels available before each fitting cutoff.
July, August, September	Three expanding-window folds: train through the preceding month and validate the next.	Choose algorithm, hyperparameters and feature set on pooled out-of-time MAE. Never randomly shuffle the series.
October	Calibrate the alert margin and perform final release rehearsal.	Refit the chosen model on January-September first. Freeze its weights; do not select a new model using October.
November-December	Locked final test: 5,856 raw target intervals before quality exclusions.	Freeze features, model, baseline and alert policy. Open once; do not tune against the result.

At a fitting boundary, exclude labels whose interval end is later than the available-data cutoff, including a boundary row if necessary. During rolling evaluation, actual readings from earlier validation/test intervals may feed later predictions only after their recorded end plus the assumed arrival delay. This reproduces live sequential forecasting; it is not permission to fit model parameters on holdout outcomes.

3.3 Freeze the advisory and the evidence

Fix the high-load threshold from January-September. Use October only to choose an alert margin from a predeclared grid, for example {0, 2, 4, 6, 8, 10} kWh. Select the smallest margin satisfying recall and precision targets; if none does, do not claim the advisory gate passes. Do not refit the regressor after margin calibration. Report all final metrics on November-December, including timestamp-convention sensitivity and paired daily error differences.

A candidate release must pass the schema and leakage checks, outperform the frozen baseline, meet high-load and runtime criteria, and carry a reproducibility manifest. An energy manager and data owner approve promotion together. Failure means keep the baseline or existing champion. Later redesign requires a new future evaluation window; repeatedly inspecting the same test set invalidates its role as an unbiased final check.

3.4 Business validation requires new evidence

Run at least four weeks of shadow operation to measure arrival delays, operator response and missed forecasts. Then, if operations approves, compare assisted and unassisted days using matched production schedules or a pre-agreed crossover pilot. Collect throughput, production mix, downtime, tariff and actual actions. Report uncertainty and confounding. The single-plant, single-year historical file supports forecasting experiments, not a causal estimate of savings or a claim of generalisation to another factory.

4 HIGH-LEVEL ARCHITECTURE DESIGN | 25%

Separate learning from operational decisions

A OFFLINE LEARNING AND RELEASE

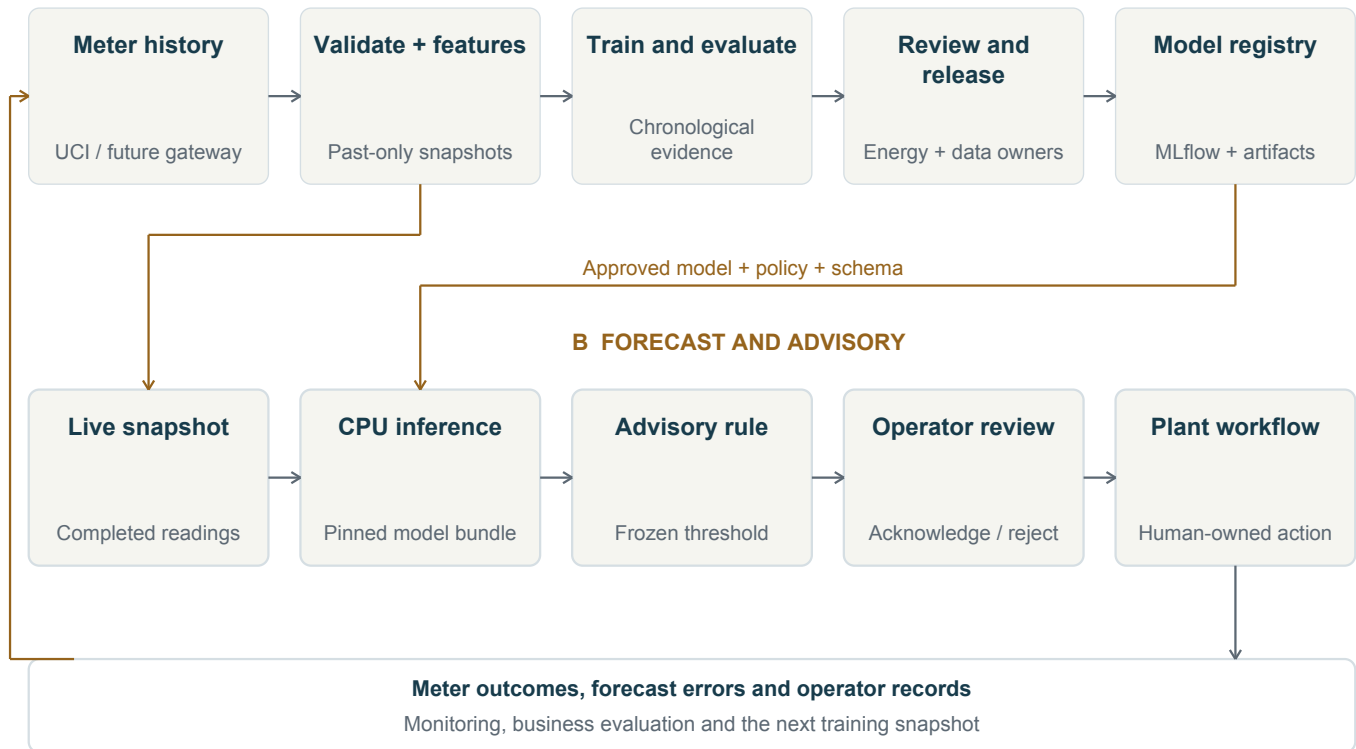


Figure 1. Proposed data and decision flow. Offline release is reviewed; online inference uses a pinned bundle. Meter outcomes return to monitoring and later training. Equipment control is outside this system.

4.1 Component responsibilities

Component	Responsibility and selected technology
Source adapter and validation	Python CSV replay for the benchmark; read-only meter API in a pilot. Enforce schema, timestamp policy, units, freshness and idempotent ingestion.
Raw and operational storage	Immutable CSV/Parquet snapshots and model artifacts in S3-compatible storage; PostgreSQL for forecast keys, arrivals, decisions and audit records.
Features and training	Shared pandas/scikit-learn transformations; past-only energy lags and calendar features. CPU Ridge and tree models are evaluated against naive baselines.
Tracking and registry	MLflow records parameters, metrics, plots, input hash, Git revision and environment. The registry holds immutable versions and a reviewed champion pointer.
Orchestration and inference	Airflow schedules replay/ingestion, weekly candidate training and outcome evaluation. A warm inference worker uses a locally cached approved model.
Operator interface and monitoring	FastAPI supplies the dashboard and acknowledgement endpoint. Structured logs, service metrics and daily accuracy reports drive alerts and human review.

4 HIGH-LEVEL ARCHITECTURE DESIGN | 25%

Lifecycle, failure recovery and lineage

4.2 Training and release flow

Ingest a versioned snapshot, validate the data, normalise timestamps under an explicit policy, then construct features with a shared library. Materialise chronological folds; fit candidate models; evaluate against the frozen baselines; register the candidate only when evidence is complete. An approved bundle includes weights, feature schema, timestamp policy, advisory parameters and dependency versions. Registering a model does not automatically authorise production use [4].

4.3 Operational flow

At each boundary t , ingestion accepts only validated readings ending no later than t . A scheduled worker builds the feature snapshot for target $[t+15, t+30)$, uses the pinned model, clips negative energy predictions to zero and applies the advisory rule. It writes a unique forecast record before notifying the UI. The operator may review a deferrable auxiliary load, acknowledge or reject the advisory; actual production actions remain in the existing operating workflow.

Airflow task retries must fit the 60-second publication budget. Permit one short retry only when enough time remains; otherwise record a missed ML deadline and run the validated fallback. Training runs in a separate worker pool so an experiment cannot delay operational forecasts. Disable historical alert emission during backfills; replay results are clearly labelled and cannot create live operator notifications [5].

Failure / signal	Response	Owner / evidence
Invalid or delayed current reading	Use same-week historical fallback only if its required row is valid; otherwise UNAVAILABLE. Show degraded status and suppress ML-based control suggestions.	IT receives a quality alert; count every missed scheduled forecast.
Model load or inference failure	Use a validated cached last-completed-reading baseline when fresh data are present. Never label fallback output as ML.	Data owner inspects bundle compatibility and can roll back.
Registry unavailable	Continue serving the last approved local bundle with checksum validation. Block new releases until registry recovery.	IT monitors cache health and release audit.
Seven-day MAE worsens by over 20% against its validation reference, or feature drift persists	Open an investigation; evaluate a candidate on a new time window. Do not auto-promote or infer that drift proves loss of accuracy.	Data owner checks labels, activity changes and data quality first.
No operator response / UI outage	Record missing acknowledgement; retain the original plant workflow. Expire an advisory when its target interval starts.	Energy manager reviews acknowledgement and incident logs.

4.4 Reproducibility and monitoring

Link each prediction to a Git commit, raw-data checksum, feature version, model version and alert-policy version. Pin dependencies, use a fixed random seed and a Docker image digest, and retain training cutoffs and machine specifications. Monitor source delay, rejection count, deadline misses, fallback rate, API latency, prediction distribution and delayed outcome error. A seed alone does not guarantee identical results across different numerical libraries or hardware.

5 TRADE-OFFS ANALYSIS | 15%

Choices, costs and reconsideration triggers

5.1 Accuracy versus simplicity and latency

A nonlinear boosting model may learn load transitions better than Ridge or persistence, but adds tuning and inspection work. Deep sequence models introduce further dependency and operational cost without evidence that this dataset needs them.

Decision: start with naive baselines, Ridge and CPU tree models; deploy the simplest candidate that passes the frozen gates. Reconsider a larger model only if residual analysis shows a repeatable gap and measured latency remains inside the publication budget.

5.2 Freshness versus implementation and operating cost

A streaming platform could react to sub-minute telemetry, while daily batch inference is cheaper but cannot support the proposed lead time. The public data itself has only 15-minute granularity.

Decision: use scheduled 15-minute inference with a warm worker and isolate weekly training. Introduce a streaming ingestion service only if live telemetry and decision needs justify it or Airflow scheduling repeatedly misses the service target. More frequent calls cannot invent finer source information.

5.3 Predictive strength versus feature availability

Contemporaneous reactive power, power factor or load category may yield attractive offline scores, but the target interval has not happened when the forecast is issued. Relying on those inputs changes forecasting into retrospective estimation.

Decision: use only historical energy and known calendar features in release one. Excluding same-interval covariates may reduce apparent accuracy but preserves a deployable contract. Add a signal only after proving its available_at timestamp and repeating out-of-time evaluation.

5.4 Alert sensitivity versus operator fatigue

A larger positive alert margin catches more high-load periods but also increases false alarms and unnecessary reviews. Optimising overall MAE does not directly settle this operational trade-off.

Decision: freeze a high-load threshold on development data and calibrate the margin on October with both recall and precision constraints. Show false alarms and missed peaks together. If no margin satisfies the constraints, keep a forecast-only pilot and redesign the advisory with the energy manager.

5.5 Automation versus production control

Automatic load shedding could act quickly but may interfere with steel quality, process sequencing and existing electrical protection. The benchmark provides no equipment constraints or action outcomes.

Decision: keep the system advisory-only and record operator decisions. Automated control requires a separately approved engineering design, plant-specific constraints and prospective evidence; a high regression score does not authorise it.

5.6 Public-data feasibility versus local validity

A licensed public dataset enables reproducible experiments immediately, but one historical plant cannot establish performance at another site or in 2026. Its midnight convention also requires an explicit assumption.

Decision: use UCI as a transparent benchmark, include timestamp sensitivity and require a prospective local shadow period. Refit and revalidate for each new site. The benefit of quick experimentation does not remove the need to verify meter semantics and operational value.

APPENDIX A IMPLEMENTATION READINESS

A concrete path into Assignment 2

Assignment 1 establishes the design. Assignment 2 can implement and evaluate this same forecasting contract with the downloaded dataset. The matrix below is a predeclared starting point, not an experiment results table. All runs use identical chronological folds and common eligible rows; preprocessing, wall-clock training time, metrics and artifacts are logged in MLflow.

Run	Configuration	Feature / tuning choice
E01	Persistence baseline	Predict target j from energy at $j-2$.
E02	Daily seasonal baseline	Predict from $j-96$.
E03	Weekly seasonal baseline	Predict from $j-672$.
E04	Ridge regression	$\alpha=1$; scale numeric features on training only.
E05	Random forest, constrained	200 trees; $\text{max_depth}=8$; $\text{min_samples_leaf}=5$.
E06	Random forest, flexible	200 trees; $\text{max_depth}=\text{None}$; $\text{min_samples_leaf}=2$.
E07	Histogram gradient boosting	Squared-error loss; $\text{learning_rate}=0.05$; 300 iterations; 15 leaves; $L2=1$.
E08	Histogram gradient boosting	Same as E07, with 31 leaves.
E09	Histogram gradient boosting	Same as E08, with $\text{learning_rate}=0.10$.
E10	Feature ablation	E08 settings, but remove rolling features and daily/weekly energy lags.

E04-E09 share calendar features, the energy lags and rolling means in Section 2.5. E10 keeps calendar information and short energy lags through $j-17$. Use $\text{random_state}=42$ where applicable, one CPU thread for controlled comparisons, and disable histogram boosting's internal random validation split ($\text{early_stopping}=\text{False}$). Apply non-negative clipping consistently when evaluating and serving regressors. These settings are proposals, not selected winners.

Expected evidence for the next assignment

Produce a results table for all 10 runs, fold-level MAE/RMSE/WAPE, high-load diagnostics, error plots and a justified candidate recommendation. Include MLflow logging/registration snippets, an Airflow DAG with dependencies and retry rules, a configuration example, type hints/docstrings, data and model versioning, a Dockerfile and reproducible run commands. The locked test is used only after development and October calibration are complete.

Artifacts already available versus future work

Available now: the original public CSV, UCI metadata and licence-page snapshot, a repeatable audit script, its JSON results and this revised design report. Still to implement: feature engineering, model experiments, MLflow tracking, Airflow DAGs, API and deployment verification. No training results or operational savings are asserted in this document.

The historical dataset is sufficient to test forecasting and pipeline behaviour. Production actions, tariffs and process constraints must be collected separately to test the business case.

REFERENCES AND ASSESSMENT COVERAGE

Sources, provenance and report map

[1] Sathishkumar V E, Shin, C., and Cho, Y. Steel Industry Energy Consumption. UCI Machine Learning Repository. DOI: 10.24432/C52G8C. Dataset page, metadata and CSV accessed 24 September 2026. Source of plant attribution, schema and public data.
<https://archive.ics.uci.edu/dataset/851/steel+industry+energy+consumption>

[2] Creative Commons. Attribution 4.0 International (CC BY 4.0). Licence identified on the UCI dataset page; attribution and a notice of modifications must accompany redistributed derivatives.
<https://creativecommons.org/licenses/by/4.0/>

[3] Hyndman, R. J., and Athanasopoulos, G. Forecasting: Principles and Practice, 3rd edition. Sections 5.8-5.10: forecast accuracy, time-series cross-validation and prediction intervals. Methodological guidance for temporal evaluation and error measures.
<https://otexts.com/fpp3/accuracy.html>

[4] MLflow documentation. Model Registry. Reference for the proposed separation of registered model versions and controlled release aliases; the exact package version will be pinned during implementation.
<https://mlflow.org/docs/latest/ml/model-registry/>

[5] Apache Airflow documentation. Best Practices. Reference for idempotent tasks, workflow implementation and operational discipline; the exact Airflow version will be pinned during implementation.
<https://airflow.apache.org/docs/apache-airflow/stable/best-practices.html>

Local evidence and transformations

The raw CSV is retained without modification. The audit can be repeated with scripts/audit_data.py; evidence/data-audit.json records verified counts and the timestamp-normalisation assumption. Downloaded metadata and the UCI licence-page HTML are preserved alongside it. Source timestamps and UCI target roles are described explicitly; this report changes the modelling task to future Usage_kWh regression.

Raw CSV SHA-256
9b1cee6f9cb9cd9df2b95814ca90a9a2ff15
b7f5f1fba0fae3c643e82072eacc
(The two lines above concatenate to one 64-character checksum.)

Assignment 1 criterion	Where addressed
Problem definition - 20%	Section 1: context, current-process assumption, measurable problem, stakeholders and ML justification.
Requirements analysis - 20%	Section 2: timing, input/output contract, functional/non-functional requirements, source, quality, privacy and volume.
Goals and metrics - 20%	Section 3: business/system/model hierarchy, definitions, thresholds, baselines and evaluation protocol.
High-level architecture - 25%	Section 4 and Figure 1: data flow, learning/inference stages, components, technologies and failure handling.
Trade-offs analysis - 15%	Section 5: six explicit decisions, drawbacks and conditions for reconsideration.